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Studierichting: Informatica
Oefeningenreeks 1C nr. 49.4

$$z^8 = -128 + 128i\sqrt{3}$$

$$r = \sqrt{(-128)^2 + (128\sqrt{3})^2} = 256$$

$$\theta = \mathrm{Bgtan}\left(\frac{128\sqrt{3}}{-128}\right) = \frac{2\pi}{3}$$

- $z_0 = \mathrm{cis}\left(\frac{\pi}{12}\right) = \frac{\sqrt{6}+\sqrt{2}}{2} + i\frac{\sqrt{6}-\sqrt{2}}{2}$
- $z_1 = \mathrm{cis}\left(\frac{\pi}{3}\right) = 1 + i\sqrt{3}$
- $z_2 = \mathrm{cis}\left(\frac{7\pi}{12}\right) = -\frac{\sqrt{6}-\sqrt{2}}{2} + i\frac{\sqrt{6}+\sqrt{2}}{2}$
- $z_3 = \mathrm{cis}\left(\frac{5\pi}{6}\right) = -\sqrt{3} + i$
- $z_4 = \mathrm{cis}\left(\frac{13\pi}{12}\right) = -\frac{\sqrt{6}+\sqrt{2}}{2} - i\frac{\sqrt{6}-\sqrt{2}}{2}$
- $z_5 = \mathrm{cis}\left(\frac{4\pi}{3}\right) = -1 - i\sqrt{3}$
- $z_6 = \mathrm{cis}\left(\frac{7\pi}{4}\right) = \frac{\sqrt{6}-\sqrt{2}}{2} - i\frac{\sqrt{6}+\sqrt{2}}{2}$
- $z_7 = \mathrm{cis}\left(\frac{11\pi}{6}\right) = \sqrt{3} - i$

References